

THE THIRTY-SECOND WILLIAM LOWELL PUTNAM MATHEMATICAL COMPETITION

December 4, 1971

- A-1. Let there be given nine lattice points (points with integral coordinates) in three dimensional Euclidean space. Show that there is a lattice point on the interior of one of the line segments joining two of these points.
- A-2. Determine all polynomials $P(x)$ such that $P(x^2 + 1) = (P(x))^2 + 1$ and $P(0) = 0$.
- A-3. The three vertices of a triangle of sides a , b , and c are lattice points and lie on a circle of radius R . Show that $abc \geq 2R$. (Lattice points are points in the Euclidean plane with integral coordinates.)
- A-4. Show that for $0 < \varepsilon < 1$ the expression $(x + y)^n (x^2 - (2 - \varepsilon)xy + y^2)$ is a polynomial with positive coefficients for n sufficiently large and integral. For $\varepsilon = .002$ find the smallest admissible value of n .
- A-5. A game of solitaire is played as follows. After each play, according to the outcome, the player receives either a or b points (a and b are positive integers with a greater than b), and his score accumulates from play to play. It has been noticed that there are thirty-five non-attainable scores and that one of these is 58. Find a and b .
- A-6. Let c be a real number such that n^c is an integer for every positive integer n . Show that c is a non-negative integer.

- B-1. Let S be a set and let $\circ$ be a binary operation on S satisfying the two laws

$$x \circ x = x \text{ for all } x \text{ in } S, \text{ and}$$

$$(x \circ y) \circ z = (y \circ z) \circ x \text{ for all } x, y, z \text{ in } S.$$

Show that $\circ$ is associative and commutative.

- B-2. Let $F(x)$ be a real valued function defined for all real x except for $x = 0$ and $x = 1$ and satisfying the functional equation $F(x) + F\{(x-1)/x\} = 1 + x$. Find all functions $F(x)$ satisfying these conditions.
- B-3. Two cars travel around a track at equal and constant speeds, each completing a lap every hour. From a common starting point, the first starts at time $t = 0$ and the second at an arbitrary later time $t = T > 0$. Prove that there is a total period of exactly one hour during the motion in which the first has completed twice as many laps as the second.

- B-4. A "spherical ellipse" with foci A, B on a given sphere is defined as the set of all points P on the sphere such that $\widehat{PA} + \widehat{PB} = \text{constant}$. Here $\widehat{PA}$ denotes the shortest distance on the sphere between P and A . Determine the entire class of real spherical ellipses which are circles.

- B-5. Show that the graphs in the x - y plane of all solutions of the system of differential equations

$$x'' + y' + 6x = 0, y'' - x' + 6y = 0 \quad (' = d/dt)$$

which satisfy $x'(0) = y'(0) = 0$ are hypocycloids, and find the radius of the fixed circle and the two possible values of the radius of the rolling circle for each such solution. (A hypocycloid is the path described by a fixed point on the circumference of a circle which rolls on the inside of a given fixed circle.)

- B-6. Let $\delta(x)$ be the greatest odd divisor of the positive integer x . Show that $\left| \sum_{n=1}^x \delta(n)/n - 2x/3 \right| < 1$, for all positive integers x .